

Unit 1 — Existence Discontinuity of Quantum Particles

Intermittent existence, intrinsic dephasing and selection of the preferred basis (HDEQ)

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Abstract

We introduce the *Hypothesis of Quantum Existence Discontinuity* (HDEQ): the active existence of a quantum system is intermittent. Within a continuous time, the evolution alternates, in cycles of characteristic duration τ_0 , between an active phase of Hamiltonian dynamics and an inactive phase of dynamical suspension. Four points are established, with particular care given to the scope of each statement. **(i)** The intermittent evolution, implemented by a controlled gating $\hat{U}(\tau_0)$ on $\mathcal{H}_{\text{ext}} = \mathcal{H}_{\text{sys}} \otimes \mathcal{H}_{\text{temp}}$, is *globally unitary* (Theorem 1, [proof]). This unitarity is *structural*: it guarantees internal consistency but by itself produces no decoherence. **(ii)** A phase-stability criterion selects, as the preferred basis, the eigenstates of the effective Hamiltonian $\hat{H}_{\text{eff}}$ (Proposition 1, [proof]); we specify the regime where this basis coincides with the einselection pointer states and the regime where it differs. **(iii)** Decoherence arises *not* from intermittency but from an extension: a dispersion $\delta\tau_0$ of the cycle durations, which acts as a parametric dephasing channel. Depending on whether $\delta\tau_0$ is quasi-static or dynamically stochastic, one obtains either a refocusable inhomogeneous dephasing (T_2^* , Gaussian) or an irreversible decoherence floor (T_2 , exponential); only the latter deserves the label « irreducible » [numerical]. **(iv)** Taken in a structurally consistent way (the effective Hamiltonian is $\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$, « prescription B »), HDEQ renders a universal factor ε *unobservable* in ratios of frequencies and rates; the relation $\Gamma_{\text{HDEQ}} = \varepsilon^2 \Gamma_{\text{QED}}$ belongs to a *distinct prescription* (A), not derivable from the postulate alone, and is not simultaneously consistent with (i)–(ii). The spectroscopic sector is therefore non-discriminating; the primary signature is the intrinsic dephasing of (iii). The existence factor ε and the period τ_0 are here phenomenological; a possible geometric origin ($\varepsilon = E(k)/K(k)$) is sketched in the conclusion. The framework does not claim to solve the measurement problem in its strong form.

Status: [proof] exact · [numerical] reproducible · [conjecture] argued · [heuristic] analogy · [speculation] · [to check] external ref. · [open] to be established.

1 Methodological note

This manuscript results from a particular methodology. The original intuition was formulated by the author around 1990; its mathematical framing was carried out in an iterative dialogue with analytic-assistance systems, under the author’s direction and supervision. Each statement carries an epistemic status label; the results labelled [proof] (Theorem 1, Proposition 1) are proved explicitly in the body (§3, §5). Final scientific validation belongs to the experimental community; full responsibility for the content rests with the author.

2 Introduction

2.1 Original intuition

In a hydrogen atom, the nucleus ($R_{\text{nuc}} \approx 1.2 \times 10^{-15}$ m) occupies, within the atomic volume ($R_{\text{at}} \approx 0.5 \times 10^{-10}$ m), a minute fraction of space: $\eta_{\text{sp}} \equiv (R_{\text{nuc}}/R_{\text{at}})^3 \approx 1.4 \times 10^{-14}$. [heuristic] By dimensional analogy: if the existence of matter is sparse along the spatial dimension, let us

examine an analogous intermittency of its *active* existence along the temporal dimension — not a granularity of time, but a dynamical activity that switches on and off within a continuous time. The role of η_{sp} is strictly heuristic, not a mathematical input of HDEQ.

2.2 Statement of the postulate

We introduce the existence factor $\varepsilon \equiv \tau_{\text{active}}/\tau_0 \in (0, 1]$, where τ_0 is the cycle duration and $\tau_{\text{active}} = \varepsilon\tau_0$ the active fraction. The complementary fraction $(1 - \varepsilon)\tau_0$ is a dynamical suspension. $\varepsilon \rightarrow 1$ recovers continuous evolution; $\varepsilon \ll 1$ corresponds to a maximally intermittent existence.

2.3 What HDEQ addresses and does not address

HDEQ proposes a structural mechanism for (a) the selection of the preferred basis and (b) an intrinsic dephasing requiring no *spatial bath*. It does not claim to derive the Born rule, nor to solve the selection of a single branch. *Framing caveat*: the decoherence of §4 arises from a stochastic dispersion of τ_0 , which is mathematically a classical noise channel; « without environment » here means « without spatial bath », not « without any source of noise ».

3 Formal framework and unitarity

3.1 Postulate and enlarged Hilbert space

Evolution proceeds in cycles of duration τ_0 : during $\varepsilon\tau_0$ the system evolves unitarily under $\hat{H}_{\text{sys}}$; during $(1 - \varepsilon)\tau_0$ its evolution is suspended. The time parameter remains continuous; it is the *action* of $\hat{H}_{\text{sys}}$ that is intermittent. We introduce

$$\mathcal{H}_{\text{temp}} = \text{span}\{|a\rangle, |i\rangle\}, \quad \mathcal{H}_{\text{ext}} = \mathcal{H}_{\text{sys}} \otimes \mathcal{H}_{\text{temp}}, \quad \hat{P} = \hat{I}_{\text{sys}} \otimes |a\rangle\langle a|, \quad \hat{Q} = \hat{I}_{\text{sys}} \otimes |i\rangle\langle i|,$$

with $\hat{P}^2 = \hat{P}$, $\hat{Q}^2 = \hat{Q}$, $\hat{P}\hat{Q} = \hat{Q}\hat{P} = 0$, $\hat{P} + \hat{Q} = \hat{I}_{\text{ext}}$. We set $\hat{U}_{\text{sys}} = \exp(-i\hat{H}_{\text{sys}}\varepsilon\tau_0/\hbar)$ and the lift $\tilde{U}_{\text{sys}} = \hat{U}_{\text{sys}} \otimes \hat{I}_{\text{temp}}$.

3.2 Theorem 1 (global unitarity)

Definition 1 (cycle propagator). $\hat{U}(\tau_0) = \tilde{U}_{\text{sys}}\hat{P} + \hat{Q} = \hat{U}_{\text{sys}} \otimes |a\rangle\langle a| + \hat{I}_{\text{sys}} \otimes |i\rangle\langle i|$: system evolution on the active branch, identity on the inactive branch (controlled gating).

Lemma 1 (tensor commutation). $[\hat{P}, \tilde{A}] = [\hat{Q}, \tilde{A}] = 0$ for $\tilde{A} = \hat{A} \otimes \hat{I}_{\text{temp}}$.

Proof. $\hat{P}$ and $\tilde{A}$ act on disjoint factors: $(\hat{A} \otimes \hat{I})(\hat{I} \otimes |a\rangle\langle a|) = \hat{A} \otimes |a\rangle\langle a| = (\hat{I} \otimes |a\rangle\langle a|)(\hat{A} \otimes \hat{I})$. $\square$

Theorem 1. [proof] $\hat{U}(\tau_0) \in \text{U}(\mathcal{H}_{\text{ext}})$.

Proof. $\hat{U}^\dagger = \hat{P}\tilde{U}_{\text{sys}}^\dagger + \hat{Q}$. Then $\hat{U}^\dagger\hat{U} = \hat{P}\tilde{U}_{\text{sys}}^\dagger\tilde{U}_{\text{sys}}\hat{P} + \hat{P}\tilde{U}_{\text{sys}}^\dagger\hat{Q} + \hat{Q}\tilde{U}_{\text{sys}}\hat{P} + \hat{Q}^2$. Term 1 equals $\hat{P}^2 = \hat{P}$ ($\tilde{U}_{\text{sys}}$ unitary); terms 2 and 3 vanish by Lemma 1 and $\hat{P}\hat{Q} = 0$; term 4 equals $\hat{Q}$. Hence $\hat{U}^\dagger\hat{U} = \hat{P} + \hat{Q} = \hat{I}_{\text{ext}}$, and symmetrically $\hat{U}\hat{U}^\dagger = \hat{I}_{\text{ext}}$. $\square$

Corollary 1 (gating and effective Hamiltonian). [proof] $\hat{U}(\tau_0)$ coincides, on the active branch, with the propagator of the square-wave Hamiltonian $\hat{H}(t) = \chi(t)\hat{H}_{\text{sys}}$ ($\chi \in \{0, 1\}$, period τ_0). For time-independent $\hat{H}_{\text{sys}}$, each cycle applies on the active branch the propagator $\exp(-i\hat{H}_{\text{sys}}\varepsilon\tau_0/\hbar)$, and composing N cycles gives

$$\hat{U}(N\tau_0) = \exp\left(-\frac{i}{\hbar}\varepsilon\hat{H}_{\text{sys}}T\right), \quad T = N\tau_0, \quad \implies \quad \boxed{\hat{H}_{\text{eff}} = \varepsilon\hat{H}_{\text{sys}}}$$

(Floquet Hamiltonian), the continuous limit being immediate. For $\hat{H}_{\text{sys}}(t)$, $\hat{H}_{\text{eff}} = \varepsilon\bar{H} + O(\tau_0)$ (Magnus expansion).

Scope. Theorem 1 is a *structural consistency guarantee*: any evolution generated by a Hermitian generator is unitary. Two consequences frame the rest of the document. **(U1)** Global unitarity implies that *no decoherence follows from intermittency alone*; the decoherence of §4 arises from an added ingredient (§4.1). **(U2)** The derivation fixes $\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$ (prescription B); this structural choice conditions the phenomenology (§7).

4 Intrinsic dephasing

4.1 Origin of the dephasing: an explicit extension

[conjecture] The computation below rests on a *phenomenological extension*: the assumption that the elementary durations vary slightly, $\tau_0^{(i)} = \tau_0 + \xi_i$, $\langle \xi_i \rangle = 0$, $\langle \xi_i \xi_j \rangle = (\delta\tau_0)^2 \delta_{ij}$. This dispersion is *not* a consequence of the fundamental postulate; averaging over $\{\xi_i\}$ is mathematically a classical dephasing channel. The physical nature of ξ_i determines the decoherence regime — a decisive point, treated below.

4.2 Two regimes, two conclusions

Let $\Delta\Phi(T) = \sum_{i=1}^N \delta\phi_i(T)$ be the relative phase accumulated between two macroscopically distinct branches. Decoherence is defined by $\text{Var}(\Delta\Phi(T_{\text{dec}})) = 1$. Model: over $T = n\tau_0$ cycles, a constituent's phase is $\phi = (\bar{E}/\hbar) \sum_j \tau_{\text{active}}^{(j)}$, sensitive to the cycle-duration dispersion $\{\xi_i\}$.

(a) Quasi-static dispersion (T_2^* , refocusable). If ξ_i is *frozen* per constituent, the phase offset is *correlated* across all cycles and grows $\propto (\bar{E}/\hbar)(\delta\tau_0/\tau_0)T$; summed over N independent constituents it yields a *quadratic* variance

$$\text{Var}(\Delta\Phi) \sim N (\bar{E}/\hbar)^2 (\delta\tau_0/\tau_0)^2 T^2, \quad T_{\text{dec}}^{-1} \propto \sqrt{N} \frac{\delta\tau_0}{\tau_0^2},$$

a *Gaussian* decay. [numerical] This regime is an *inhomogeneous* dephasing, in principle *refocusable by echo* (Hahn-type): it therefore does *not* constitute an irreducible floor.

(b) Dynamic dispersion (T_2 , irreducible). If ξ_i *re-fluctuates* cycle by cycle (a stochastic process), the phase offset performs a *random walk* of step $(\bar{E}/\hbar)\delta\tau_0$ over $n = T/\tau_0$ cycles (variance $\propto n$), so the variance becomes *linear* in T ,

$$\text{Var}(\Delta\Phi) \sim N (\bar{E}/\hbar)^2 (\delta\tau_0)^2 \frac{T}{\tau_0}, \quad T_{\text{dec}}^{-1} \propto N \frac{(\delta\tau_0)^2}{\tau_0^3},$$

an *exponential* decay, *not* refocusable. [numerical] *Only this regime* deserves the label « irreducible ». The prefactor depends on the characteristic energy $\bar{E}$ (taken $\sim \hbar/\tau_0$ for the estimate); the precise value remains [open].

Distinction from environmental decoherence. In both cases, the prefactor depends only on *intrinsic* parameters $(\delta\tau_0, \tau_0, N)$, not on an external bath; a system ideally isolated from its surroundings would still dephase. But HDEQ is a noise model (intrinsic), conceptually close to spontaneous-collapse models (§9). Note the *N-exponent*: $\sqrt{N}$ (static) or N (dynamic), to be compared with the often linear or super-linear scaling of collisional models — a point to exploit for experimental discrimination (§7).

5 Selection of the preferred basis

Definition 2 (survival amplitude). For $|\psi\rangle \in \text{Dom}(\hat{H}_{\text{eff}})$, $\|\psi\| = 1$: $A_\psi(t) = \langle\psi|e^{-i\hat{H}_{\text{eff}}t/\hbar}|\psi\rangle$. The state $|\psi\rangle$ is a *pointer state* (stable phase) iff $|A_\psi(t)| = 1$ for all t .

Proposition 1. [proof] $|A_\psi(t)| = 1 \forall t \iff |\psi\rangle$ is an eigenstate of $\hat{H}_{\text{eff}}$.

Proof. ($\Leftarrow$) immediate. ($\Rightarrow$) by Cauchy–Schwarz, $|A_\psi(t)| \leq \|\psi\| \|e^{-i\hat{H}_{\text{eff}}t/\hbar}\psi\| = 1$, with equality iff $e^{-i\hat{H}_{\text{eff}}t/\hbar}|\psi\rangle = \lambda(t)|\psi\rangle$, $|\lambda(t)| = 1$. Since $|\psi\rangle \in \text{Dom}(\hat{H}_{\text{eff}})$, Stone’s theorem gives differentiability at $t = 0$: $-\frac{i}{\hbar}\hat{H}_{\text{eff}}|\psi\rangle = \lambda'(0)|\psi\rangle$, i.e. $\hat{H}_{\text{eff}}|\psi\rangle = (i\hbar\lambda'(0))|\psi\rangle$ (real eigenvalue). $\square$

Corollary 2 (modulus defect at order τ_0^2). [proof] The short-time expansion $A_\psi(t) = 1 - \frac{i}{\hbar}\langle\hat{H}_{\text{eff}}\rangle t - \frac{1}{2\hbar^2}\langle\hat{H}_{\text{eff}}^2\rangle t^2 + O(t^3)$ gives $|A_\psi(t)|^2 = 1 - (\Delta H_{\text{eff}}/\hbar)^2 t^2 + O(t^4)$, with $(\Delta H_{\text{eff}})^2 = \langle\hat{H}_{\text{eff}}^2\rangle - \langle\hat{H}_{\text{eff}}\rangle^2$. The defect vanishes iff $|\psi\rangle$ is an eigenstate, coinciding with the short-time Zeno bound.

Scope. The criterion selects $\text{Spec}(\hat{H}_{\text{eff}})$. **(1)** With $\hat{H}_{\text{eff}} = \varepsilon\hat{H}_{\text{sys}}$, this is the *energy* basis of $\hat{H}_{\text{sys}}$. If $\hat{H}_{\text{eff}} = \hat{H}_S + \hat{H}_A + \hat{H}_{SA}$ includes a *strong* measurement coupling, its eigenstates approach the interaction basis, hence the einselection pointer states; in the *free/weak* regime, the energy basis *differs* from them (quasi-localised basis). Selection is therefore *regime-dependent* [conjecture]. **(2)** Zeno-type mechanism: the gating stabilises the eigenbasis of the generator. **(3)** The Proposition identifies a *basis*, not the selection of a single outcome nor the Born rule.

6 The Born rule as a heuristic postulate

[open] A rigorous derivation of the Born rule from a dynamical framework remains an open problem for all known frameworks (Gleason assumes the measure-theoretic structure; Deutsch–Wallace rationality axioms; Zurek an entanglement structure). HDEQ does not break this impasse: the frequency of outcomes is taken as a heuristic postulate. A scope analysis sharpens this status (§10.4): Born is inherited from standard canonical quantization, not derived from the scale sector.

7 Experimental directions

7.1 Direction I: intrinsic dephasing by interferometry

[numerical] The *primary* signature of the framework. Its nature depends on the regime of §4: an *irreducible* floor requires a *dynamic* dispersion (case b, exponential decay, $T_{\text{dec}}^{-1} \propto N(\delta\tau_0)^2/\tau_0^3$); the static case (a) is in principle cancellable by echo and does not constitute a fundamental test. Molecular interferometry ($\sim 25\,000$ amu today; targeted extension 10^6 – 10^8 amu in cryogenic ultra-vacuum) must *separate the N -scaling* of HDEQ ($\sqrt{N}$ or N depending on regime) from that of the residual noise, at fixed N and with an independent calibration of Γ_{env} .

7.2 Direction II: atomic transitions — non-discriminating under the consistent framework

[open] *Two gating prescriptions.* **(A)** gate only the interaction $\hat{H}_{\text{int}}$ (free evolution maintained): the golden rule gives $\Gamma = |\langle f|\varepsilon\hat{H}_{\text{int}}|i\rangle|^2\rho = \varepsilon^2\Gamma_{\text{QED}}$, frequencies unchanged. **(B)** gate the full $\hat{H}_{\text{sys}}$: $\hat{H}_{\text{eff}} = \varepsilon\hat{H}_{\text{sys}}$ rescales *all* energies.

- The fundamental postulate, properly derived (Cor. 1), *is* prescription B. Prescription A requires an *additional ingredient* (selective gating of the interaction) not contained in the single postulate.

- Under B, every ratio of frequencies or rates equals $\nu_1/\nu_2 = \omega_1/\omega_2$: ε *cancels*. A universal ε is therefore *unobservable* in comparative spectroscopy, including by the best clocks (uncertainty 3×10^{-18}). The spectroscopic sector is non-discriminating.
- The relation $\Gamma = \varepsilon^2 \Gamma_{\text{QED}}$ and the central relation $\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$ are *not simultaneously consistent*: they belong to different prescriptions. Adopting B (structural consistency) renders Direction II *null*; adopting A makes it a prediction *conditional* on an additional postulate.

Conclusion: unless prescription A is independently motivated, atomic spectroscopy does not constrain HDEQ. This is consistent with the non-observability of the clocks. The E3 clock transition of $^{171}\text{Yb}^+$ (sub-Hz) would in any case be unusable.

7.3 Direction III (exploratory): stroboscopic resonances

[speculation] By analogy with Floquet ($\Omega_F = \varepsilon \Omega_R$), periodic structures could appear at $\Delta E = 2\pi\hbar/\tau_0$. Amplitude, boost-invariant transcription and confrontation with data remain open.

8 Extension to quantum field theory

[proof] Let $\mathcal{F}_{\text{ext}} = \mathcal{F}_{\text{sys}} \otimes \mathcal{H}_{\text{temp}}$. The creation/annihilation operators commute with the projectors ($[\hat{P}, \hat{a}(p)] = 0$, $\hat{P}$ acting as the identity on the system part); the equal-time canonical relations are preserved. *Ontological identification* [conjecture]: the inactive phases *could* be identified with vacuum fluctuations; this reading is distinct from the level-I formalism, which remains valid independently. *UV scope* [proof]: HDEQ carries no dynamical UV cutoff at $\Lambda_{\text{UV}} = \hbar/\tau_0$ — such a cutoff would be untenable (LHAASO/GRB 221009A $E_{\text{QG},1} \gtrsim 10 E_{\text{Planck}}$ [to check]; electron and muon $g-2$). Level I (unitarity, preferred basis, dephasing) is independent of this question.

9 Positioning: NCG, CSL, Zeno, Trotter

Non-commutative geometry. Connes’ programme and the Connes–Rovelli modular flow share with HDEQ the rejection of continuous time as a primary datum and decoherence without a spatial bath; $\hat{P}(\tau_0)$ could be interpreted as a discrete sampling of the Tomita–Takesaki modular automorphism [conjecture]. **Collapse models (CSL/GRW).** The dispersion-induced decoherence of §4 is mathematically a stochastic dephasing channel, close to CSL *without jumps*: HDEQ preserves global unitarity (on $\mathcal{H}_{\text{ext}}$) where CSL adds non-unitary noise on $\mathcal{H}_{\text{sys}}$. Distinction to be made explicit [conjecture]. **Zeno dynamics.** Proposition 1 is a Zeno-type stabilisation (the eigensubspace of the generator is stabilised by the repeated gating). **Trotter.** The gated formalism is formally a Trotter evolution with an « idle » step; the specificity of HDEQ is the *ontological interpretation* (intermittent active existence) and the dispersion $\delta\tau_0$, not the time slicing itself [heuristic].

10 Discussion, limits and scope

10.1 Scope regarding the unification question

HDEQ establishes the mechanism of the *quantum sector* — intermittent existence, intrinsic dephasing, preferred-basis selection — and provides the existence factor ε and the gating structure. It does *not*, by itself, bear on the question « is gravitation quantum? ». In the framework where the metric $g_{\mu\nu}$ remains a *classical* field (scalar–tensor action in $\frac{1}{2}fR$, $\Omega = \sqrt{f}$ relating the Jordan and Einstein frames), the gravity $\leftrightarrow$ quantum coupling enters through the modulation $\tau_0(R) = 2\pi\sqrt{\omega/(\alpha + \frac{1}{2}\beta M_{\text{Pl}}^2 R)}$, whose dynamical effect $\delta\tau_0/\tau_0 = -(\beta/4)M_{\text{Pl}}^2 R/\alpha$ is unobservable

in matter ($\sim 10^{-81}$) and becomes decisive only at the critical curvature $R_c = 2\alpha/|\beta|$. The treatment of this question belongs to a later *synthesis* level, where all sectors are in scope; the metric remaining classical there, the framework takes a *falsifiable* position (absence of gravitationally induced entanglement). The present work is a *quantum prerequisite* for it, not an answer. [open]

10.2 Honest limitations

- τ_0 and ε are phenomenological, not derived from first principles (a geometric origin is conjectured, cf. §10.3).
- The dispersion ξ_i and the Gaussian hypothesis are explicit layers; the « irreducible » character requires the *dynamic* regime (§4b).
- Theorem 1 is structural: it does not produce decoherence.
- The preferred-basis criterion is regime-dependent (§5).
- The selection of a single branch is not resolved; the Born rule is heuristic (cf. scope, §10.4).
- The spectroscopic sector is non-discriminating under the consistent prescription (B).

10.3 Origin of ε and τ_0 : a geometric direction

[conjecture] HDEQ leaves ε, τ_0 free. A geometric origin is conjectured: the oscillation of a compactified scale field χ , read as a Page–Wootters relational clock, would generate the gating and fix

$$\varepsilon = \langle \cos^2(\chi/2) \rangle = \frac{E(k)}{K(k)}, \quad k = \sin(\chi_{\max}/2), \quad \tau_0 = 2\pi\sqrt{\omega/\alpha} \approx 3.5 \times 10^{-37} \text{ s.}$$

[proof] If this origin is retained, the identity derives exactly. Homogeneous reduction ($\chi = \chi(t)$): $\mathcal{L}_\chi = \frac{1}{2}\omega\dot{\chi}^2 - \alpha(1 - \cos\chi)$; Euler–Lagrange gives $\omega\ddot{\chi} + \alpha\sin\chi = 0$, and the first integral (with $\dot{\chi} = 0$ at $\chi_{\max}$) $\dot{\chi} = \sqrt{2\alpha/\omega} \sqrt{\cos\chi - \cos\chi_{\max}}$. With $k = \sin(\chi_{\max}/2)$ and the substitution $\sin(\chi/2) = k \sin\varphi$,

$$dt = \sqrt{\frac{\omega}{\alpha}} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}, \quad \tau_0 = 4\sqrt{\frac{\omega}{\alpha}} K(k), \quad \varepsilon = \frac{\int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} d\varphi}{\int_0^{\pi/2} d\varphi / \sqrt{1 - k^2 \sin^2 \varphi}} = \frac{E(k)}{K(k)},$$

the last equality since $\cos^2(\chi/2) = 1 - k^2 \sin^2 \varphi$ (the $\sqrt{\omega/\alpha}$ cancel). The pendulum/elliptic-integral machinery is classical [17]. At small amplitude ($k \rightarrow 0$): $\tau_0 \rightarrow 2\pi\sqrt{\omega/\alpha}$ and $\varepsilon \rightarrow 1 - \chi_{\max}^2/8$; at the separatrix ($\chi_{\max} \rightarrow \pi \equiv \infty^*$, $k \rightarrow 1$): $K \rightarrow \infty$, $\varepsilon \rightarrow 0$ (suspension). At the vacuum $\varepsilon \rightarrow 1$, rendering intermittent effects unobservable.

10.4 Scope of the quantum sector: a geometric mechanism of time built on QM

[proof] HDEQ level-I *inherits* the Born rule from its standard canonical quantization; it neither derives it nor departs from it. Once the quantum sector is a complex Hilbert space endowed with unitary dynamics (Theorem 1), the usual structural derivations (Gleason [5]; Masanes–Galley–Müller [14]) apply. The scale sector, by contrast, is entirely *real*: the field χ is a real angle on S_χ^1 governed by a pendulum Lagrangian, $\mathcal{L}_\chi = -\frac{1}{2}\omega(\partial\chi)^2 - \alpha(1 - \cos\chi)$, and the existence factor $\varepsilon = \langle \cos^2(\chi/2) \rangle = E(k)/K(k)$ is a real time-average there [proof]. The imaginary unit enters only *after* the geometry, at two points: at the quantization of χ (Mathieu/Schrödinger equation) and at the Page–Wootters relational passage, $\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$, which presupposes an already-quantum clock.

That this boundary is one of principle, not a gap, rests on two independent arguments. (i) *Internal*: the inter-sector symmetry $\mathbb{Z}_2 \times \mathbb{Z}_2$ is discrete; Noether’s theorem does not apply, so no

bilinear current $j^0 \sim \psi^* \psi$ follows (the current associated with the periodicity of the real angle is linear in $\partial\chi$, a scale momentum). (ii) *External*: under the standard composition rule, a real formulation of quantum theory makes predictions distinct from the complex one, which experiment settles in favour of the complex [15, 16]; the imaginary unit is therefore empirically required, not extractable from a real Lagrangian alone.

Scope statement. Scale closure explains the clock, relational time, and the existence factor ε — all real and derived. It does *not* explain the origin of the complex structure. **HDEQ is thus a geometric mechanism of time built on quantum mechanics, and not a derivation of quantum mechanics from scale closure.**

10.5 Conclusion

HDEQ proposes a mechanistic framework for the structural selection of the preferred basis and for an intrinsic dephasing, preserving global unitarity by construction and compatible with current phenomenological constraints. Its claims are calibrated: unitarity is structural; « irreducible » decoherence requires a dynamic dispersion; atomic spectroscopy is non-discriminating under the consistent prescription. The primary testable signature is the intrinsic dephasing in mass interferometry. The origin of ε, τ_0 could be carried by a compactified-scale geometry — a direction left to later development, under the scope made precise in §10.4.

Symbol index. ε existence factor; τ_0 cycle duration; $\hat{P}, \hat{Q}$ active/inactive projectors; $\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$ effective Hamiltonian; $\delta\tau_0$ intrinsic dispersion; N number of constituents; $\chi(t)$ gating; Γ transition rate.

Status labels. The results labelled [proof] (Theorem 1, Corollary 1, Proposition 1) are proved in the body. The estimates of §4 are [numerical]/[open]. The external references marked [to check] (LHAASO bounds, recent $g - 2$ values) must undergo a final bibliographic verification before deposit.

Reproducibility

The [numerical] statements and the symbolic verification of the [proof] results are reproduced by a standalone script (NumPy/SciPy), with sections coded so as to be specific to this document:

- [H1] Theorem 1: verification of global unitarity $\hat{U}^\dagger \hat{U} = \hat{I}_{\text{ext}}$ and of the composition $\hat{U}(N\tau_0) = e^{-i\varepsilon \hat{H}_{\text{sys}} N\tau_0}$ on the active branch ($\hat{H}_{\text{eff}} = \varepsilon \hat{H}_{\text{sys}}$).
- [H2] Proposition 1: survival amplitude $|A_\psi(t)| = 1$ for an eigenstate, < 1 otherwise, and agreement of the t^2 coefficient with $(\Delta H_{\text{eff}})^2$.
- [H3] Dephasing: Monte-Carlo of the two regimes — T -exponent equal to 2 (quasi-static, Gaussian, refocusable) and 1 (dynamic, exponential, irreducible), variance $\propto N$ in both cases.
- [H4] Constants: $\eta_{\text{sp}}, \varepsilon = E(k)/K(k)$ (with $\varepsilon \rightarrow 1$ as $k \rightarrow 0$), scales $\hbar/\tau_0$ and $2\pi\hbar/\tau_0$.

File: `U1_HDEQ_reproducible.py`. *Convention*: $\hbar = \tau_0 = 1$ unless otherwise stated. The prefix `U1_HDEQ_` and the [H·] codes are specific to this document, to avoid any collision with other script sets.

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